

Mario Espinoza-Gutierrez

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Summary

Computer Science student and aspiring full-stack developer who builds complete products, from research and UX through a tested backend and a typed front end. Currently shipping mX, a personal AI assistant with a FastAPI backend and React + TypeScript client, and Mira, a React Native check-in app for older and disabled adults. Comfortable owning a project end to end and learning whatever the next layer of the stack requires.

Education

Indiana University Indianapolis — Indianapolis, IN · Jan 2024 – May 2028 (expected)

B.S. in Computer Science · Full-Stack Development Certificate · Honors College

Relevant coursework:

- **Computer science:** Intro to Computers & Programming (C200), Intro to Software Systems (C212), Discrete Structures (C241), Data Structures in Python (C310), Computer Structures (C335)
- **Web & media development:** Intro to Front-End Development (N115), Intermediate Front-End Development (N215), Intro to Media Application Development (N220), Dynamic Data Applications (N322), Design Issues in Digital Media (N201)
- **Human-computer interaction:** HCI Principles & Practice (I270)
- **Math:** Linear Algebra for Data Science (I243), Calculus for Data Science I (I241)

Projects

[mX](#) — Personal AI assistant (Jarvis-style) · Python, FastAPI, React, TypeScript, SQLite

- Built a full-stack AI assistant with a FastAPI backend, password and API-key auth, 30-day sessions, and SQLite chat history
- Designed a provider interface with model adapters so the assistant can swap hosted LLM APIs without touching app code

- Built the chat client in React 19 + TypeScript with Vite, rendering Markdown, code highlighting, and LaTeX math
- Tested both layers with pytest and Vitest + Testing Library, tracked coverage, and wrote versioned prompts with an eval suite
- Containerized with Docker and prepared Fly.io deployment config; one-command local launcher for Windows
- Roadmap: memory and personalization, tool integrations, and a hands-free AR glasses interface

[Mira](#) — Daily check-in app for older and disabled adults · React Native, Expo

- Designing a zero-friction check-in product: join in under 60 seconds by phone, a named caregiver, and a daily pulse via SMS or call
- Ran user research and usability tests on two clickable prototypes before implementation; Prototype v2 approved
- Building the React Native app with AI coding agents, keeping voice/phone and press-1 flows first-class so older users need no install

Maro — Programming language built from scratch · Java

- Wrote a tree-walk interpreter (~2,500 lines, 9 files) with an interactive REPL and 21 built-in functions
- Presented the language design and implementation in an academic setting

Responsive web projects — [TechFlow](#), [Trail & Timber](#) · HTML, SCSS, JavaScript

- Built multi-page responsive sites with modular Sass partials, centralized media queries, and CSS-only mobile navigation
- Used Playwright-based visual rendering to catch layout bugs across screen sizes

Experience

Undergraduate Researcher, 1RIP Research Program — Indiana University Indianapolis · Aug 2025 – Aug 2026

- Worked with Prof. Juexin Wang on a grant-funded project applying optimal transport methods to single-cell RNA sequencing data
- Integrated Harmony batch correction into a MOSCOT analysis pipeline in Python
- Presented results as a poster at a university library research symposium and prepared the project's submission package

Utility Locator — USIC · Indianapolis, IN · Feb 2023 – Jul 2025

- Located and marked underground electric, gas, telecom, water, and sewer lines to prevent excavation damage
- Documented every job with photos and records, and coordinated directly with excavators on site
- Worked outdoors in all conditions while upholding public-safety standards

Technical Skills

- **Front end:** React, React Native, TypeScript, JavaScript (ES6+), HTML5, CSS3, Sass/SCSS, responsive design, Vite
- **Back end:** Python, FastAPI, REST APIs, authentication and sessions, SQLite, LLM API integration
- **Languages:** Python, JavaScript, TypeScript, Java, C++, C, C#
- **Testing:** pytest, Vitest, React Testing Library, Playwright, code coverage
- **Tools & DevOps:** Git, GitHub, Docker, Fly.io, Expo, Linux, GDB, AI coding agents

Interests

- **Building products end to end:** I like taking an idea from user research to a deployed app, and I'm growing mX and Mira into full products.
- **Linux and systems:** I run Linux daily and built a system with Linux From Scratch; that low-level view makes me a better back-end and DevOps developer.
- **Interactive web and games:** I build browser games and 3D experiments with React and Three.js, and hackathon apps such as a React fitness tracker.

References

Juexin Wang, Faculty Mentor (1RIP), Indiana University Indianapolis · wangjuex@iu.edu